TASK 1_2 REPORT

Technical Analysis Trading System

1. TECHNICAL INDICATOR

Simple Moving Average (SMA)

- **Mathematical Formula:**

$$\mathrm{SMA}_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

Where P(t−i) is the Adjusted Close price at time t−i, and n is the parameters lookback window (n=20,50).

- **Measurement Objective:** The SMA serves as an arithmetic smoothing filter designed to isolate the primary direction of a market by stripping out localized, high-frequency transactional noise.
- **Lag Characteristics: Pure Lagging.** Because it assigns equal mathematical weight to every data point within the lookback window, its reaction time is delayed by exactly n/2 intervals, meaning it cannot anticipate pivot points ahead of price action.
- **Optimal Operating Environment:** Highly reliable in persistent, multi-month, institutional trending markets. Conversely, it is highly prone to producing lagging false signals in highly mean-reverting or tightly consolidating market structures.

Exponential Moving Average (EMA)

- **Mathematical Formula:**

$$\mathrm{EMA}_t = (P_t \times \alpha) + (\mathrm{EMA}_{t-1} \times (1 - \alpha)), \quad \alpha = \frac{2}{n+1}$$

Initialized with an arithmetic mean background anchor, computed with `adjust=False`.

- **Measurement Objective:** Like the SMA, the EMA tracks directional trends. However, it applies a geometric smoothing multiplier that heavily weights the most recent price points.

- **Lag Characteristics: Lagging with Reduced Latency.** While still tracking past data, the exponential decay weighting allows the EMA to react to sudden momentum shifts significantly faster than an SMA of the identical lookback length.
- **Optimal Operating Environment:** Best suited for fast-moving, accelerating trends and volatile price breakouts. Its core risk is heightened vulnerability to sharp, temporary intraday counter-trend spikes ("whipsaws").

Relative Strength Index (RSI)

- **Mathematical Formula:**

$$\text{RSI}_t = 100 - \left(\frac{100}{1 + \text{RS}_t} \right), \quad \text{RS}_t = \frac{\text{Smoothed Gain}_{t,14}}{\text{Smoothed Loss}_{t,14}}$$

Calculated using Wilder's technique where:

$$\text{Smoothed Gain}_t = \frac{(\text{Smoothed Gain}_{t-1} \times 13) + \text{Gain}_t}{14}$$

- **Measurement Objective:** A bounded momentum oscillator varying between 0 and 100 that quantifies the velocity and behavioral overextension of price changes.
- **Lag Characteristics: Leading / Coincident.** Rather than following trailing trend lines, it identifies when buying or selling pressure has reached unsustainable statistical extremes before those reversals show up in trend lines.
- **Optimal Operating Environment:** Highly effective in range-bound, horizontal, or cyclical consolidation phases where the established thresholds (>70 Overbought, <30 Oversold) are well-respected. It can become highly deceptive during strong, runaway trends, where it will lock into overextended states for months at a time.

Moving Average Convergence Divergence (MACD)

- **Mathematical Formula:**

$$\text{MACD Line}_t = \text{EMA}(12)_t - \text{EMA}(26)_t$$

$$\text{Signal Line}_t = \text{EMA}(\text{MACD Line}, 9)_t$$

$$\text{Histogram}_t = \text{MACD Line}_t - \text{Signal Line}_t$$

- **Measurement Objective:** Captures the shifting relationship between moving averages to evaluate changes in trend strength, direction, momentum, and structural duration.

- **Lag Characteristics: Coincident Momentum, Lagging Signals.** The standalone MACD line responds directly to immediate shifts in the price spread, while the Signal Line crossover lag is governed by the secondary 9-period smoothing wrapper.
- **Optimal Operating Environment:** Highly effective in wide, liquid, cyclical macro swings. It struggles in compressed sideways ranges, where the histogram will constantly fluctuate across the zero-line, generating false entry triggers.

Bollinger Bands

- **Mathematical Formula:**

$$\text{Middle Band}_t = \text{SMA}(20)_t$$

$$\text{Upper / Lower Bands}_t = \text{SMA}(20)_t \pm (2 \times \sigma_{t,20})$$

- **Measurement Objective:** Maps relative overbought/oversold boundaries while dynamically adjusting to changing market volatility regimes.
- **Lag Characteristics: Coincident Volatility, Lagging Centerline.** The outer bands expand or contract immediately based on rolling statistical variance, while the directional core remains bound to the lagging 20-period SMA.
- **Optimal Operating Environment:** Excellent for breakout identification during compressed volatility phases (a "band squeeze") and for execution filtering when trading mean-reversion ranges.

2. BACKTESTING METHODOLOGY & VALIDATION CONTROL

To construct a realistic historical profile, the backtesting engine separates signal logic from order execution, eliminating look-ahead bias and tracking all transaction frictions.

Signal Isolation and Look-Ahead Bias Prevention

Look-ahead bias is a common flaw in technical backtests that happens when a model executes trades at the day t closing price using indicators calculated from that same close. This assumes the trader can execute an order at the exact millisecond the session ends.

To eliminate this bias, this engine enforces a strict time-shifted separation:

1. Indicators are computed at the **Close of Day t**, when all market data for that session is locked and final.
2. If a crossover condition is met, a signal is recorded (+1 or 0).
3. The signal vector is shifted forward by exactly one period (`df['Signal'].shift(1)`).

4. The trading loop executes the trade exclusively at the **Open price of Day t+1**.

This ensures that no future data affects execution pricing, mirroring the exact conditions of live systematic trading.

Capital Allocation & Transaction Cost Frictions

- **Initial Capital Anchor:** Fixed at exactly **Rs. 1,00,000** at the launch of each backtest.
- **Frictional Execution Costs:** A mandatory **0.1% transaction cost** is levied against every single trade leg (0.2% round-trip friction).
    - **On Buy Orders:** Net Capital Spent=Cash Balancet×(1−0.001).
    - **On Sell Orders:** Cash Received=(Shares Held×Open Pricet+1)×(1−0.001).

Simulation Limitations

- **Slippage Assumptions:** The engine assumes perfect fills at the reported historical Open price. In live trading, large order sizes or thin market liquidity can create adverse slippage, resulting in execution prices worse than the posted open.
- **Liquidity and Order Matching:** The simulation assumes infinite market liquidity, meaning any size position can be instantly liquidated at the exact open quote without moving the market price against the strategy.
- **Intraday Volatility Blindness:** Portfolio values are tracked and updated using daily closing prices. This means the model does not monitor or report maximum intraday paper losses (adverse excursions) that occur within a single trading session.

3. EMPIRICAL PERFORMANCE ANALYSIS

The backtesting framework evaluated the primary **EMA (20/50)** strategy across three distinct market profiles over a historical horizon.

Core Performance Metrics Summary Table

Asset Name	Strategy CAGR	B&H CAGR	Strategy Sharpe	B&H Sharpe	Strategy Max Drawdown	B&H Max Drawdown	Strategy Win Rate
M&M.NS	19.15%	31.56%	0.840	1.115	-31.09%	-28.09%	41.67%
ONGC.NS	20.92%	30.79%	0.625	1.022	-28.28%	-35.30%	63.64%
HDFCBANK.NS	-6.65%	0.78%	-0.321	0.142	-37.63%	-27.78%	38.46%

In-Depth Asset Profiles

1. M&M.NS The Strong Uptrend

- **Performance Summary:** The strategy delivered a solid **19.15% CAGR** and a **0.840 Sharpe Ratio** across 12 completed round trips, though it trailed behind the exceptionally strong Buy-and-Hold index (**31.56% CAGR**).
- **Behavioral Dynamics:** `M&M.NS` maintained a powerful structural uptrend over the historical window. The EMA strategy capitalized on this move by holding major trend expansions for long periods—most notably from **May 2023 to November 2023** and **November 2023 to November 2024**, which grew strategy capital from Rs. 1,23,812 to Rs. 2,59,647.
- **Underperformance Rationale:** Despite these gains, the strategy lagged Buy-and-Hold due to the natural entry lag built into moving averages and a choppy consolidation phase in **late 2024 (November 2024 to February 2025)**. During this choppy window, the system gave back profits over four consecutive losing trades, cutting capital from Rs. 259,647 down to Rs. 203,249.

2. ONGC.NS The Volatile Cyclical Outperformance

- **Performance Summary:** The strategy posted a **20.92% CAGR** with a **0.625 Sharpe Ratio** and achieved the highest accuracy with a **63.64% Win Rate**.
- **Behavioral Dynamics:** While the strategy's CAGR was lower than Buy-and-Hold (30.79%), it delivered an important structural advantage: **significant risk reduction**. It limited Maximum Drawdown to **-28.28%**, compared to the steep **-35.30%** decline suffered by buy-and-hold investors.
- **Drawdown Mitigation Analysis:** During a major cyclical downturn in late 2024, the strategy triggered a defensive exit on **September 17, 2024, at Rs. 294.65**, locking in a portfolio value of Rs. 2,54,181. The engine then safely held this capital in cash for four months while the underlying stock fell. This preserved capital and allowed the strategy to re-enter the market at a much lower price (**Rs. 265.80 on January 21, 2025**), successfully bypassing the worst of the market correction.

3. HDFCBANK.NS The Longest and Deepest Drawdown

- **Performance Summary:** The strategy failed completely on this asset, posting a negative **-6.65% CAGR**, a negative **-0.321 Sharpe Ratio**, a deep **-37.63% Max Drawdown**, and a low **38.46% Win Rate**.
- **Drawdown Analysis:** `HDFCBANK.NS` entered a long, deep drawdown starting in **April 2022** that lasted through **mid-2024**. During this time, the stock traded within a flat, horizontal consolidation range. Because trend-following strategies require clear directional paths, the EMAs frequently crossed back and forth within this range.
- **Whipsaw Erosion:** Between **April 2022 and August 2023**, the engine was trapped in a classic "whipsaw" cycle, triggering 7 consecutive losing round trips that steadily eroded capital from Rs. 96,173 down to Rs. 77,209. The lagging nature of the EMAs caused the

strategy to buy near localized price peaks and sell near local price bottoms. This continuous capital drag was further worsened by the 0.1% execution fee per leg, illustrating the danger of running trend-following systems in sideways markets.

4. PARAMETER SENSITIVITY TESTING & OVERFITTING RISK

To evaluate the robustness of our strategy rules, we performed sensitivity analysis across three different historical EMA window configurations on `M&M.NS`.

EMA Window Sensitivity Matrix (`M&M.NS`)

Configuration Window	CAGR (%)	Sharpe Ratio	Maximum Drawdown (%)
EMA (10 / 30)	16.34%	0.764	-28.77%
EMA (20 / 50) Base	19.15%	0.840	-31.09%
EMA (15 / 45)	18.15%	0.816	-32.17%

Analysis of Settings

The standard **EMA (20/50)** window combination delivered the strongest risk-adjusted performance, generating a **19.15% CAGR** and a **0.840 Sharpe Ratio**.

- The faster **EMA (10/30)** setup reduced the Maximum Drawdown to **-28.77%** due to its faster exits, but gave up overall return (**16.34% CAGR**) because it over-traded during temporary price consolidations.
- The **EMA (15/45)** arrangement underperformed both options, showing that intermediate parameters can suffer from poor timing synchronization during key market turning points.

Overfitting Risk (Data-Mining Bias)

Selecting a specific parameter combination (like 20/50) simply because it produced the highest return over a single past period introduces significant **overfitting risk (data-mining bias)**. Overfitting occurs when a mathematical system is tuned too closely to the historical noise of a specific dataset.

This danger is clearly visible when looking at the sensitivity matrix for `HDFCBANK.NS`:

- **EMA (10/30):** CAGR: -2.98% | Sharpe: -0.062 | MDD: -32.01%
- **EMA (20/50):** CAGR: -6.65% | Sharpe: -0.321 | MDD: -37.63%
- **EMA (15/45):** CAGR: -5.49% | Sharpe: -0.245 | MDD: -39.01%

On HDFC Bank, the 10/30 setting appears superior purely because its faster parameters happened to align better with that stock's specific price fluctuations during that exact multi-year period. Relying blindly on these optimized historical settings often leads to poor performance in live trading, as future market patterns are unlikely to mirror past noise.

5. RISK, FRICTIONS, & STRATEGIC LIMITATIONS

Frictional Drag Analysis (The 0.2% Round-Trip Cost)

While a 0.1% per-leg execution cost seems small, its compounding effect erodes net returns over time, especially for active trading strategies.

For example, on `HDFCBANK.NS`, the strategy executed **13 completed round trips (26 separate fee events)**.

1. **Direct Cost Impact:** Direct execution fees removed a raw **2.60%** from the overall portfolio capital.
2. **Compounding Opportunity Loss:** Because fees are deducted immediately from cash at each turn, they permanently reduce the capital available for compound growth on subsequent trades. For HDFC Bank, this ongoing frictional drag was a key factor in turning a flat baseline market return (**0.78% B&H CAGR**) into a painful loss (**-6.65% Strategy CAGR**).

Trend-Following Vulnerabilities in Sideways Markets

The underlying math of trend-following indicators requires positive auto-correlation in price action (the statistical probability that tomorrow's price move will continue in the same direction as today's). In a horizontal, range-bound market, price auto-correlation turns negative.

When an asset consolidates sideways, prices flip direction whenever they approach the boundaries of the range. A trend crossover strategy will systematically trigger buy orders near the top of the range (just as buying momentum exhausts) and sell orders near the bottom of the range (exactly when selling pressure dries up). This structural limitation locks the strategy into a cycle of consecutive losses, demonstrating that trend-following models require active directional trends to cover their inherent operational costs.

6. CONCLUSION AND STRATEGIC RECOMMENDATIONS

1. **Integrate Volatility Regime Filters:** Do not run trend-following strategies as always-on models. The system should monitor an index-wide trend metric, such as the 14-day Average Directional Index (ADX) or a rolling Efficiency Ratio. The trading engine should automatically pause trend-following strategies whenever the ADX falls below 20, shifting capital safely into cash or mean-reversion strategies.

2. **Add Volume Confirmation Rules:** To minimize false signals during early breakout attempts, require daily trading volume to exceed its 20-day rolling average by at least 1.5x before allowing an EMA crossover entry signal to execute.
3. **Expand Out-of-Sample Validation:** To avoid overfitting, lookback parameters should be optimized across a broad, diversified index of 50+ liquid assets spanning multiple market sectors. This ensures that parameter selections are chosen for structural stability across varied market environments rather than historical coincidence.